

Notebook

February 17, 2021

What might we want to investigate further? Write a few sentences below and be prepared to discuss in next week's small group meeting.

Why does Cristiano have many sources? Also, he happens to tweet from Android and iPhone. Is one an Android tablet? And the other his iPhone? Are there also other people managing his account, such as WhoSay and MobioInsider?

0.0.1 Question 2f

We just looked at the top 5 most commonly used devices for each user. However, we used the number of tweets as a measure, when it might be better to compare these distributions by comparing *proportions* of tweets. Why might proportions of tweets be better measures than numbers of tweets?

Proportions allow us to see the differences between total usage of each individual, rather than by their total tweets. We want to normalize their usage and be able to compare their frequency between each device platform.

Compare Cristiano's distribution with those of AOC and Elon Musk. In particular, compare the distributions before and after hour 6. What differences did you notice? What might be a possible cause of that? Do the data plotted above seem reasonable?

Cristiano's tweets are very normal behavior, no tweets during bedtime, but more tweets as the day progresses, hitting a peak near 6pm. As for AOC, it's the opposite. Maybe she is a night owl and sleeps during the day? But, we should notice that we didn't do any timezone correction. As for Elon Musk, his tweets are all over the place. Might be plausible, as he is known to do crazy things such as sleep near his factory and follow a "20hr work per day" mantra. These distributions might be more understandable if we apply a time zone correction.

0.0.2 Question 4f

When grouping by mentions and aggregating the polarity of the tweets, what aggregation function should we use? What might be some drawbacks of using the mean?

We should use the aggregation function `mean()`. It makes the most sense as it represents the scores, but some drawbacks might be that polarity is qualitative not quantitative.

0.0.3 Question 5a

Use this space to put your EDA code.

```
[48]: for tweet in tweets:
        twitter_stat = tweets[tweet]
        twitter_stat['length'] = twitter_stat['clean_text'].apply(lambda x: len(x))
        twitter_stat.plot(x='length', y='favorite_count', kind='scatter',
        ↳title=tweet + " User's Tweet Length to Favorites Ratio")

    for tweet in tweets:
        twitter_stat = tweets[tweet]
        twitter_stat['length'] = twitter_stat['clean_text'].apply(lambda x: len(x))
        twitter_stat.plot(x='length', y='retweet_count', kind='scatter',
        ↳title=tweet + " User's Tweet Length to Retweets Ratio")
```

c argument looks like a single numeric RGB or RGBA sequence, which should be avoided as value-mapping will have precedence in case its length matches with *x* & *y*. Please use the *color* keyword-argument or provide a 2-D array with a single row if you intend to specify the same RGB or RGBA value for all points.

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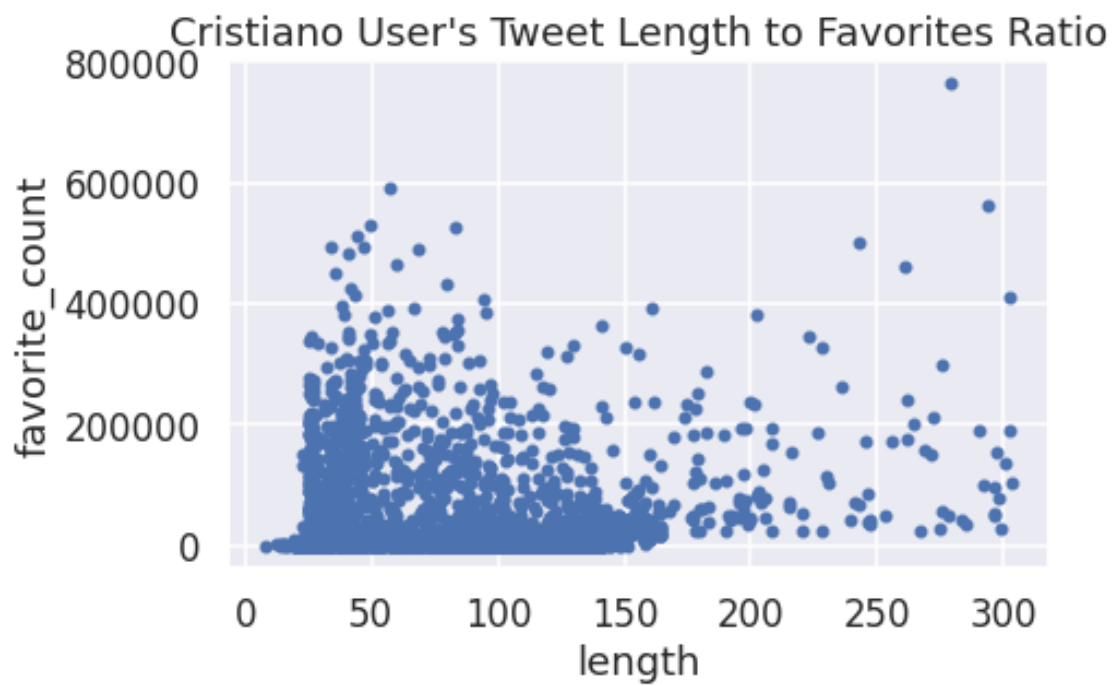
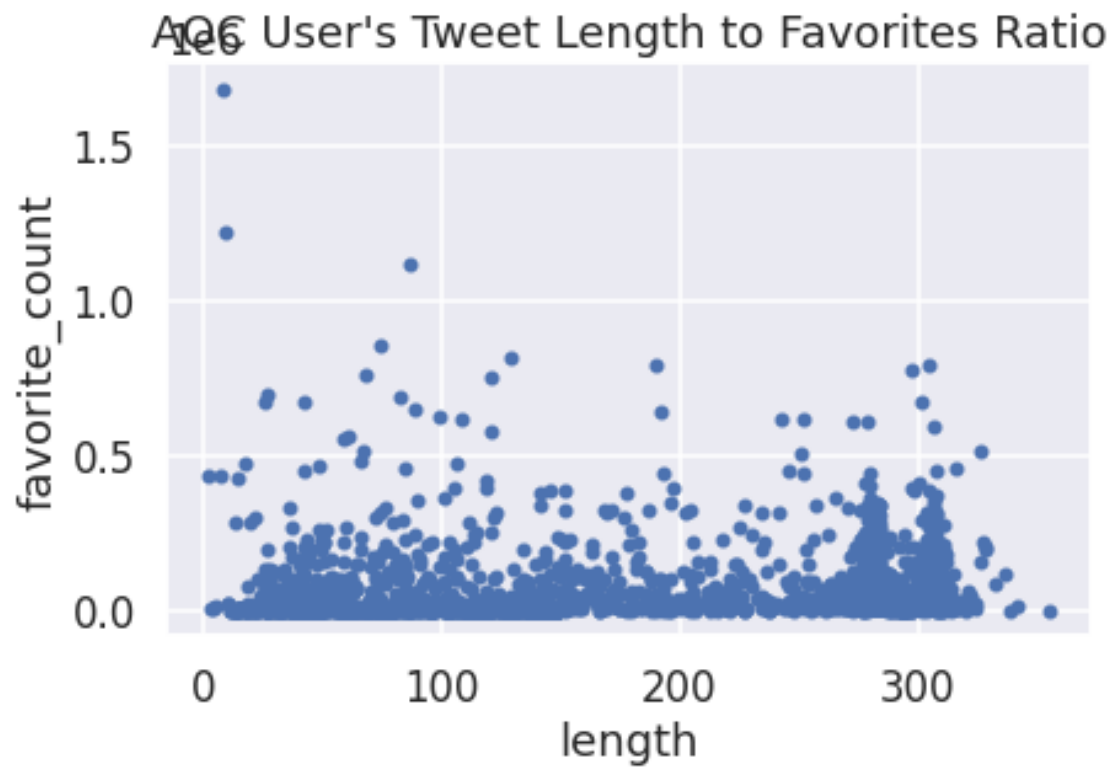
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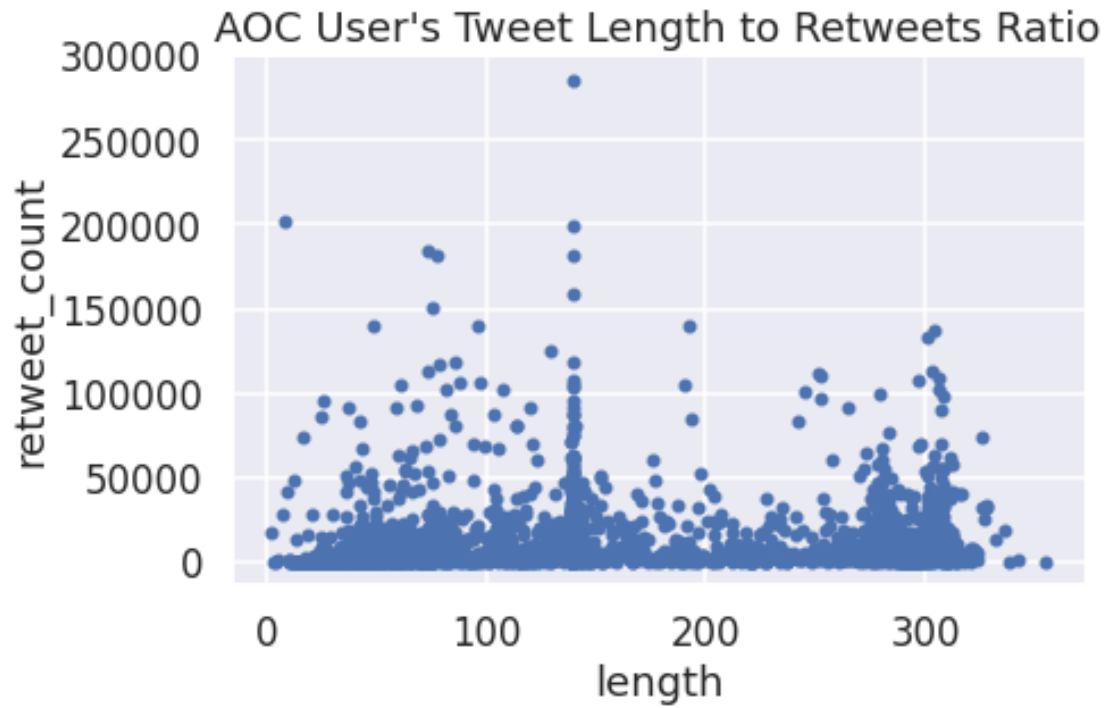
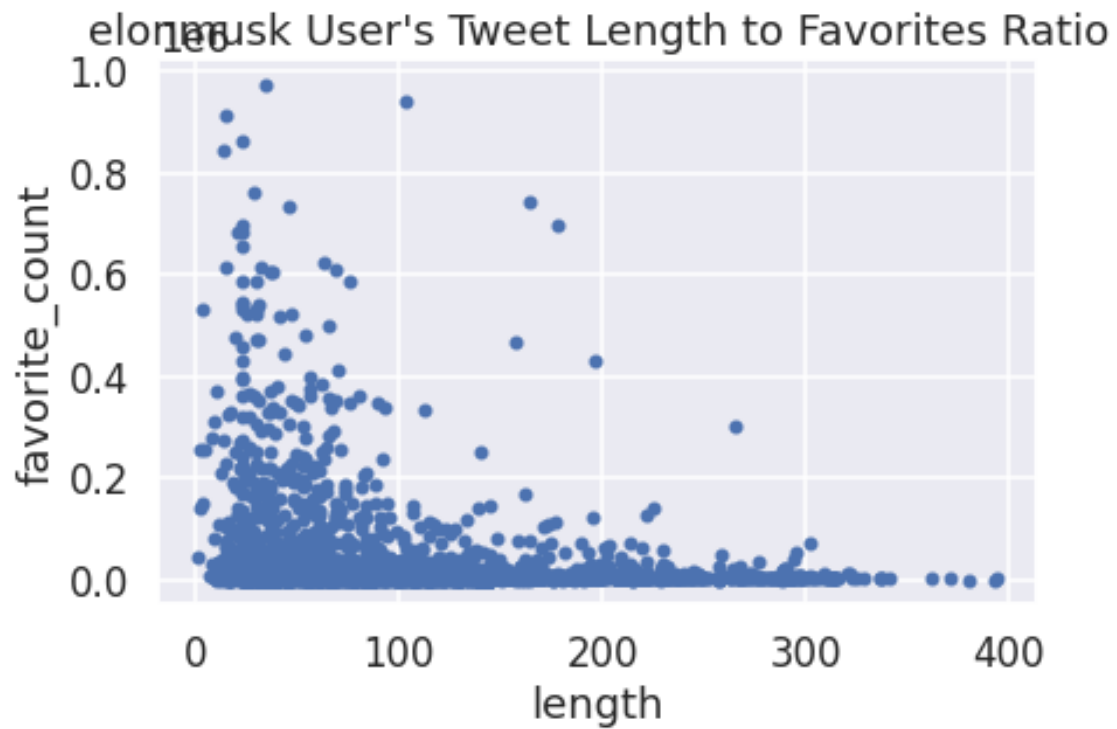
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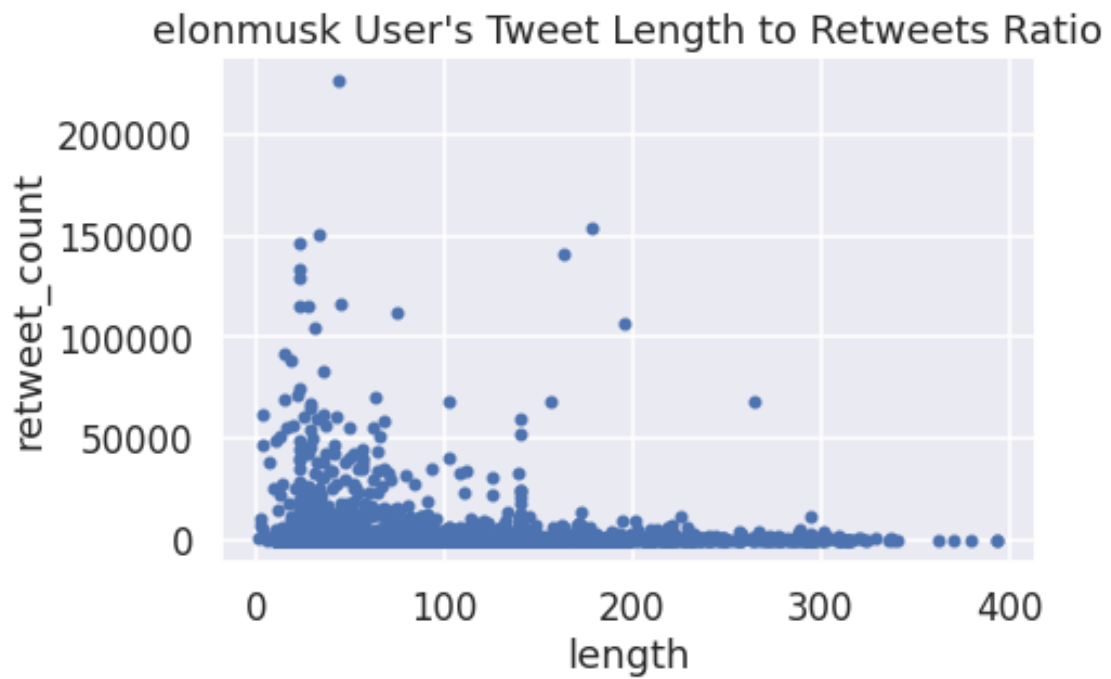
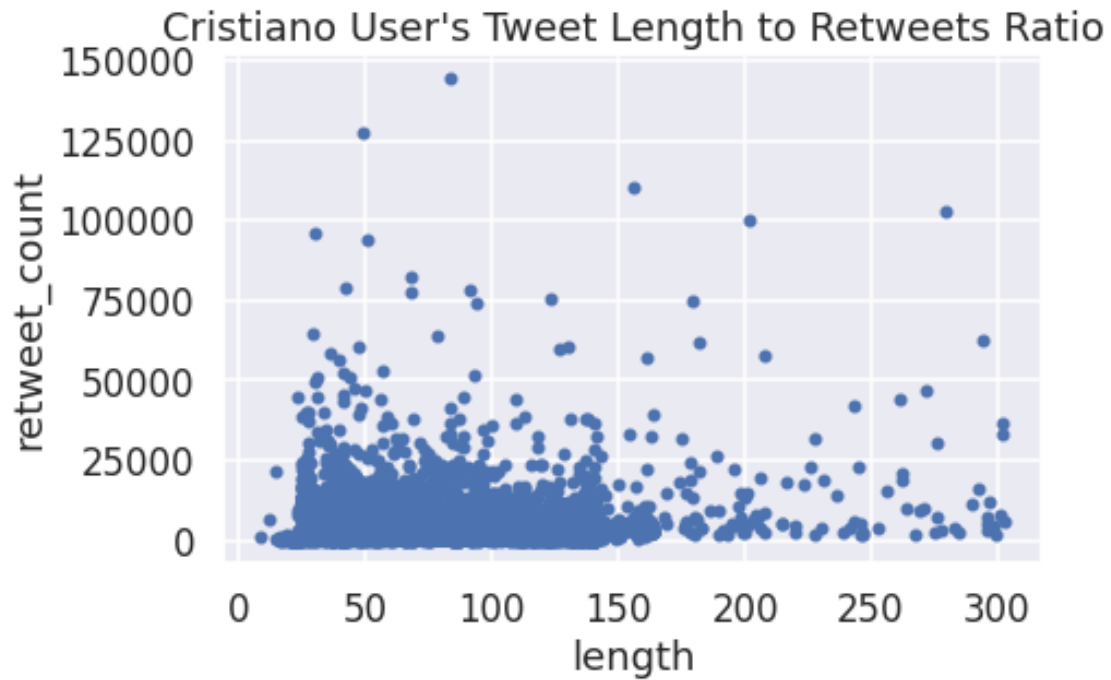
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0.0.4 Question 5b

Use this space to put your EDA description.

I wanted to see if there is any correlation between the length of the tweet and the amount of favorites/retweets it gets! Seems like a lot of people tend to react to shorter tweets. Maybe they are images or memes, or they are oneliner jokes that are funny, as shown for Elon Musk's Tweets. There are also anomalies, such as longer tweets. These tend to get more favorites, as the message might be meaningful or important. AOC has a specific anomaly at 140 characters, so maybe those are consistent tweets such as updates or automated messages. Overall, there are visual trends to recognize, as well as not so obvious ones.

Note that for some of the graphs, they are on a scale of 1 because the units are in scientific (too many favorites). Graph is hard to read for those large cases as the unit conversion is overlayed by the long title text